



Stability and Integrity of Aerospace Structures

Project deliverable II



Project II:

Wing-box upper panel sizing considering stability

Load cases:

- **Pull-up** ($n_z = 2.5$)

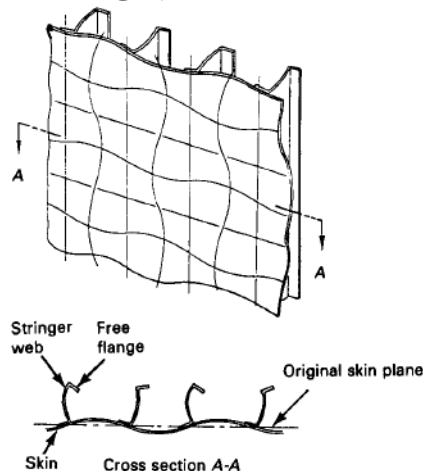
Tasks:

1. Preliminaries: geometry and loads calculation (from Project deliverable I)
2. Geometry simplifications
3. Material selections
4. Stringer shape selection
5. Design criterion (minimum weight)
6. Sizing

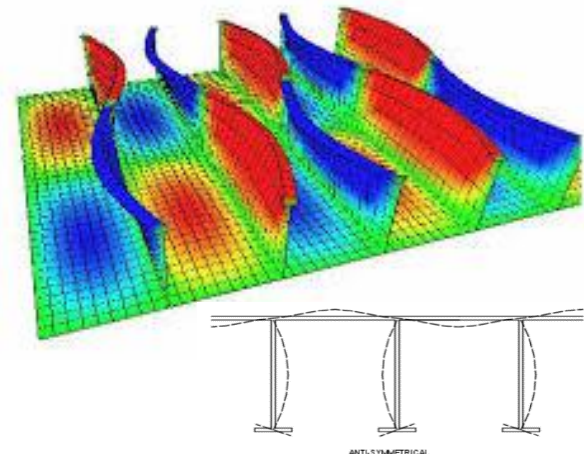
Project II: Theoretical background

- Stiffened panels can develop several separate types of instability
- Different types of buckling may be coupled to different degrees
 - Skin buckling (Initial buckling):
 - Waving of the skin between stringers (half-wavelength comparable with the stringer pitch)
 - Stringer waving induced by the skin instability
 - Local instability:
 - Short wavelength buckling involving stringer web and flange displacements with half-wavelengths comparable with the stringer depth
 - Skin movement and waving induced by the stringer instability

Skin buckling (Initial Buckling)



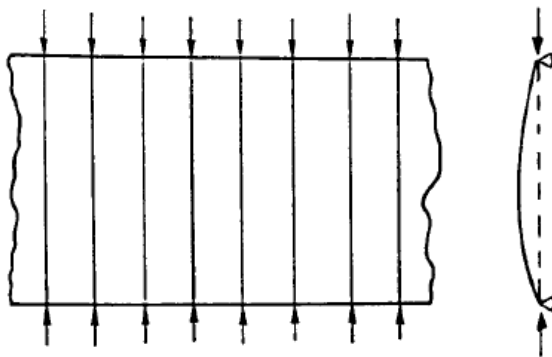
Stiffener local instability



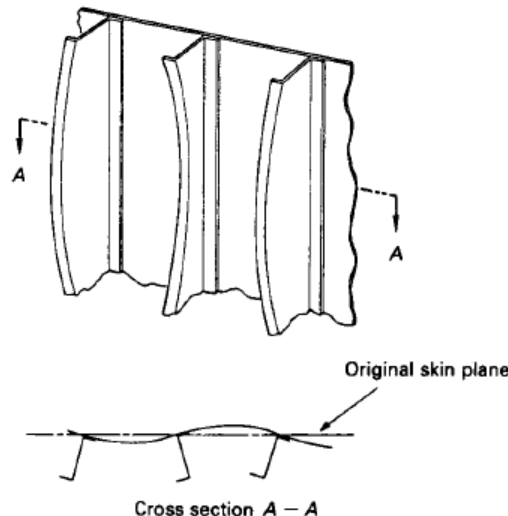
Project II: Theoretical background

- Flexural instability:
 - Simple skin/stringer combination Eulerian buckling (deflection normal to the skin plane)
 - Half-wavelength equals to rib spacing
 - Associated stringer twisting for non symmetrical section
- Torsional instability:
 - Stringer solid rotation about their longitudinal axis
 - Small associated skin distortion
 - Half-wavelength usually of the order of 3 stringer pitches

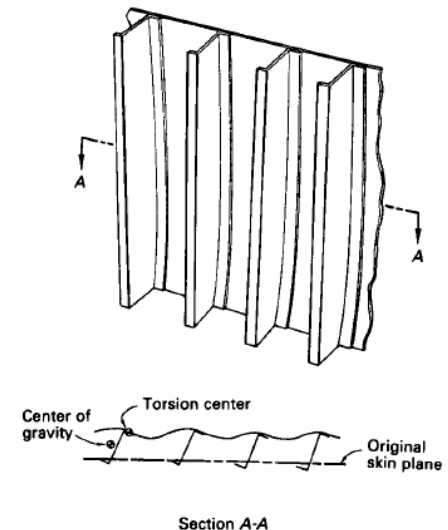
Flexural instability
(Eulerian buckling)



Torsional instability



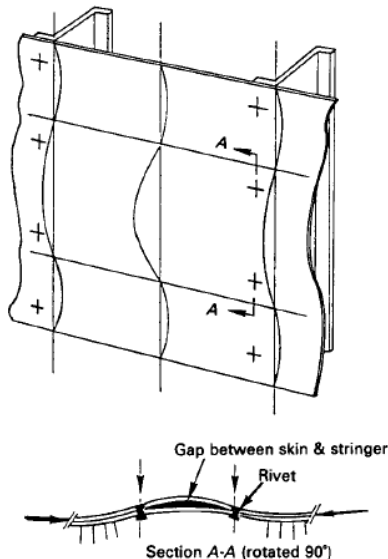
Flexural + torsional instability



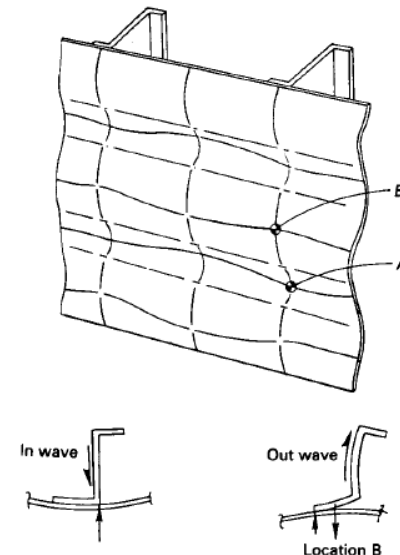
Project II: Theoretical background

- Inter-rivet buckling:
 - Skin buckling as a column between rivets
 - Constraint rivet pitch
- Wrinkling:
 - Skin short-wavelength buckling as elastic supported structures
 - Instability similar to inter-rivet buckling
 - Can be avoided keeping the rivets or attachment line close to stringer webs

Inter-rivet buckling



Wrinkling



Project II: Tasks

Tasks:

1. Preliminaries:
 - I. Retrieve the lightest wing box from Project I, and the limit condition (pull-up)
 - II. Modify the geometry to have a rectangular wing box
 - III. Material selection
2. Minimum weight sizing of stiffened panels:
 - I. Z-shaped stringers
 - II. Integral stringers
3. Minimum weight sizing of stiffened panels (manufacturing constraints):
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 - II. Integral stringers
4. Additional considerations
 - I. Material selections
 - II. Rib optimal spacing

Project II: Tasks

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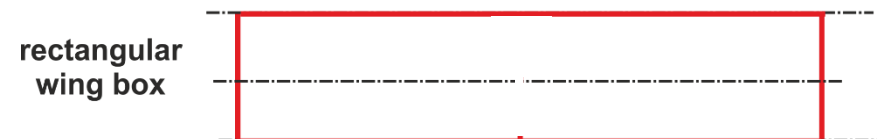
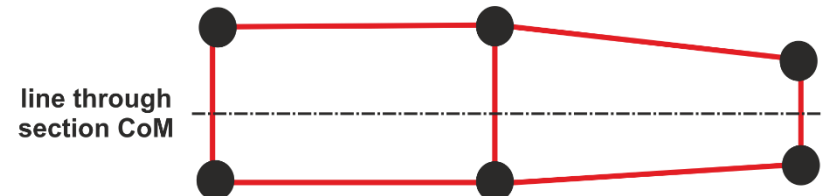
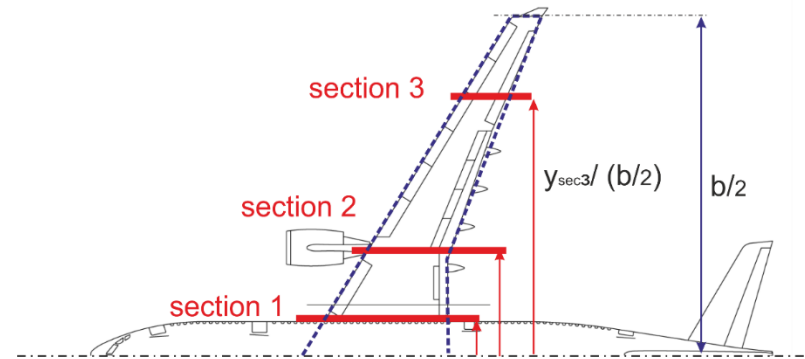
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Project II:

Task 1 - Preliminaries

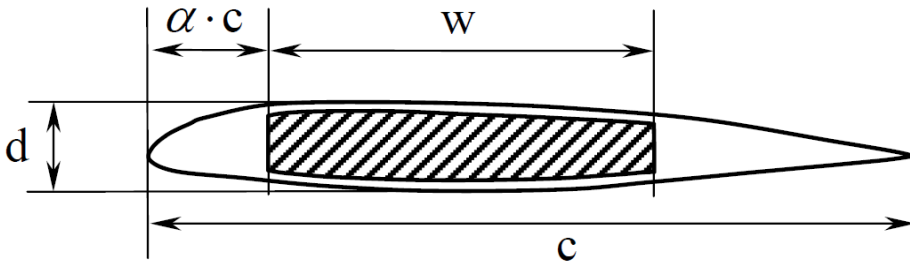
I. Retrieve the lightest wing box from Project I, and the limit condition (pull-up)

- You have all the geometrical data of the lightest wing box you found in project I
- Consider the 3 wing-box sections (root, kink and tip)
- To simplify calculations, consider an equivalent rectangular wing box
- Just consider a single cell (conventional wing box design)

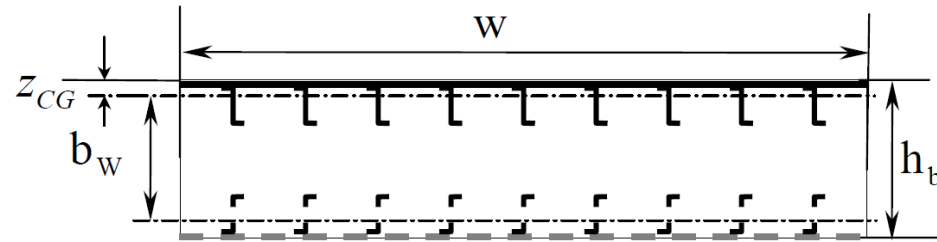


Task 1 – Wing box simplification

Real wing box section



Wing box section simplification



- Rectangular symmetric cross section (equal upper and lower panels)
- Neglect stress variation within the panel

$$J = 2 \cdot (w \cdot \bar{t}) \cdot \left(\frac{b_w}{2} \right)^2 \quad \bar{t} = t + \frac{A_s}{b}$$

$$h_b = 0.85 \cdot d$$

$$b_w = h_b - 2 \cdot z_{CG}$$

$$N = \sigma \cdot \bar{t} = \frac{M}{J} \cdot \frac{b_w}{2} \cdot \bar{t} = \frac{M}{w \cdot b_w}$$

A_s = stringer cross section area

t = skin thickness

b = stringer pitch

w = wing box width

d = max airfoil thickness

z_{CG} = panel centre of gravity coordinate

M = bending moment

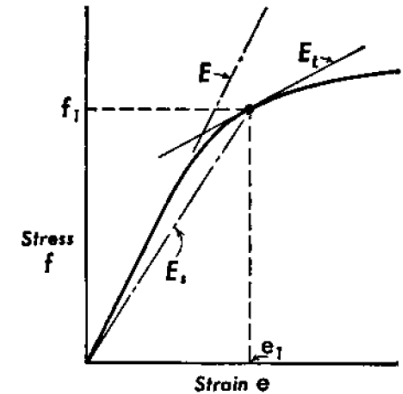
N = axial load per unit width

Task 1 – Material modelling

- Ramberg-Osgood analytical stress strain curve relation

$$\frac{E\varepsilon}{\sigma_{0.7}} = \frac{\sigma}{\sigma_{0.7}} + \left(\frac{\sigma}{\sigma_{0.7}} \right)^n$$

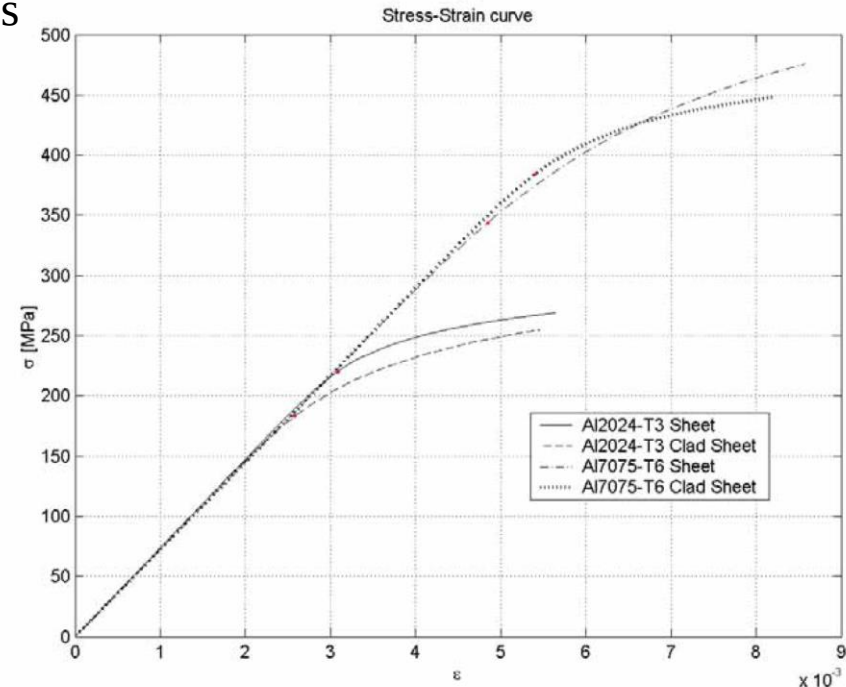
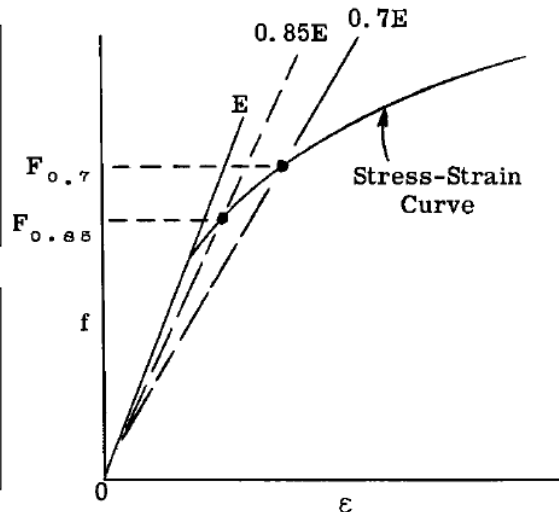
$$n = 1 + \frac{\ln\left(\frac{17}{7}\right)}{\ln\left(\frac{\sigma_{0.7}}{\sigma_{0.85}}\right)}$$



- $\sigma_{0.7}$ = stress at the intersection between the curves and $0.7E$
- $\sigma_{0.85}$ = stress at the intersection between the curves and $0.85E$
- n = hardening exponent

$$E_t(\sigma) = \frac{E}{1 + \frac{3}{7} \cdot n \cdot \left(\frac{\sigma}{\sigma_{0.7}} \right)^{n-1}}$$

$$E_s(\sigma) = \frac{E}{1 + \frac{3}{7} \cdot \left(\frac{\sigma}{\sigma_{0.7}} \right)^{n-1}}$$





Material properties

Common aerospace materials 1/2

	2024-T3 Bare-Sheet	2024-T3 Clad-Sheet	2024-T3 Extrusion	7050-T7451 Bare-Plate	2090-T83 (Al-Li)	
σ_{tu}	441	414	393	510	517	[MPa]
σ_{ty}	324	303	290	448	483	[MPa]
σ_{cy}	269	255	234	414	434	[MPa]
σ_p	220	183	183	353	374	[MPa]
$\sigma_{0.7}$	264	246	226	418	437	[MPa]
n	15	9	12	19	20	
E	72395	72395	74463	71016	79290	[MPa]
E_c	73774	73774	75842	73084	81358	[MPa]
ν	0.33	0.33	0.33	0.33	0.34	
γ	$2.78 \cdot 10^3$	$2.78 \cdot 10^3$	$2.78 \cdot 10^3$	$2.83 \cdot 10^3$	$2.59 \cdot 10^3$	[Kg/ m^3]

Material properties

- Common aerospace materials 2/2
- Range values function of the heat treatment

	7075-T6 Bare-Sheet	7075-T6 Clad-Sheet	7075- T73 Extrusion	Ti-6Al-4V Extr.Anneal.	Ti-6Al-4V Sheet aged	
σ_{tu}	538	517	469	896/1014	896/1034	[MPa]
σ_{ty}	483	455	400	827/917	827/965	[MPa]
σ_{cy}	475	448	400	855/979	855/1000	[MPa]
σ_p	343	384	359	741/849	746/872	[MPa]
$\sigma_{0.7}$	496	455	402	874/1007	875/1030	[MPa]
n	9.2	19.5	27	21	22	
E	71016	71016	71705	118590	113074	
E_c	72395	72395	73774	118590	113074	[MPa]
ν	0.33	0.33	0.33	0.31	0.31	
γ	$2.81 \cdot 10^3$	$2.81 \cdot 10^3$	$2.81 \cdot 10^3$	$4.43 \cdot 10^3$	$4.43 \cdot 10^3$	[Kg/ m ³]

Material properties

■ Mechanical properties definition

σ_{tu} = ultimate tensile strength

σ_{ty} = tensile yield stress

σ_{cy} = compressive yield stress

σ_p = elastic proportional limit

$\sigma_{0.7}$ = Ramberg-Osgood stress at $0.7E$

E = Young's modulus

E_c = compressive Young's modulus

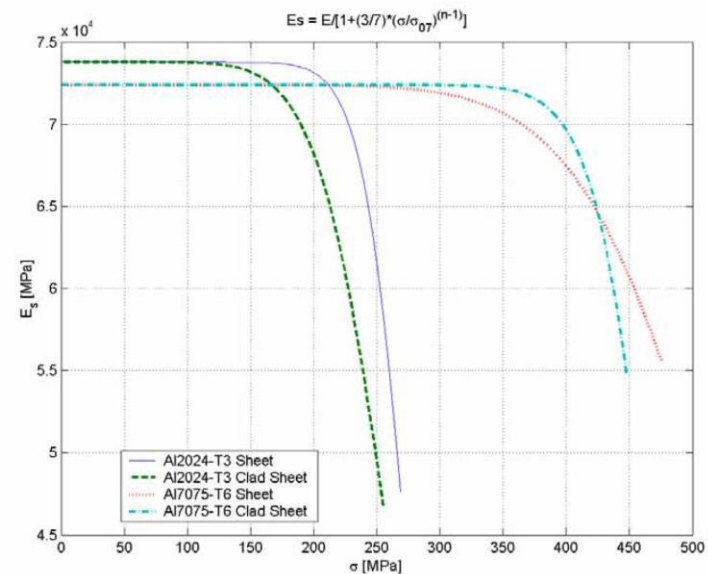
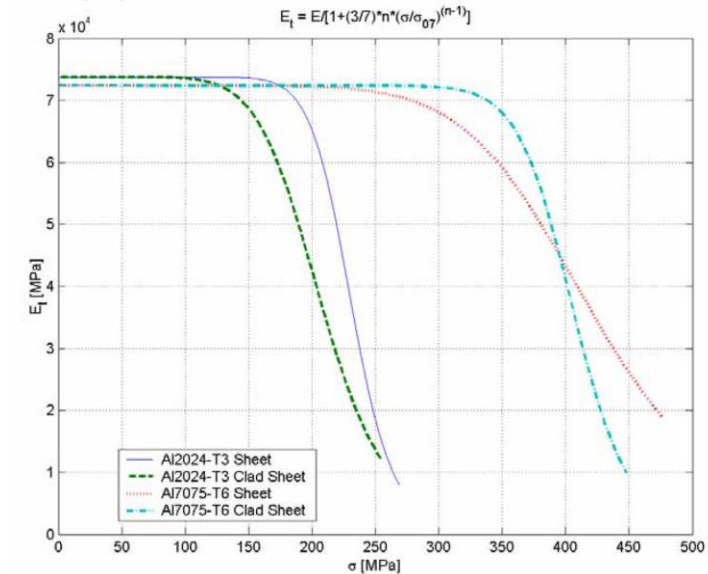
E_t = tangent modulus (Plate plastic correction)

E_s = Secant modulus (Column plastic correction)

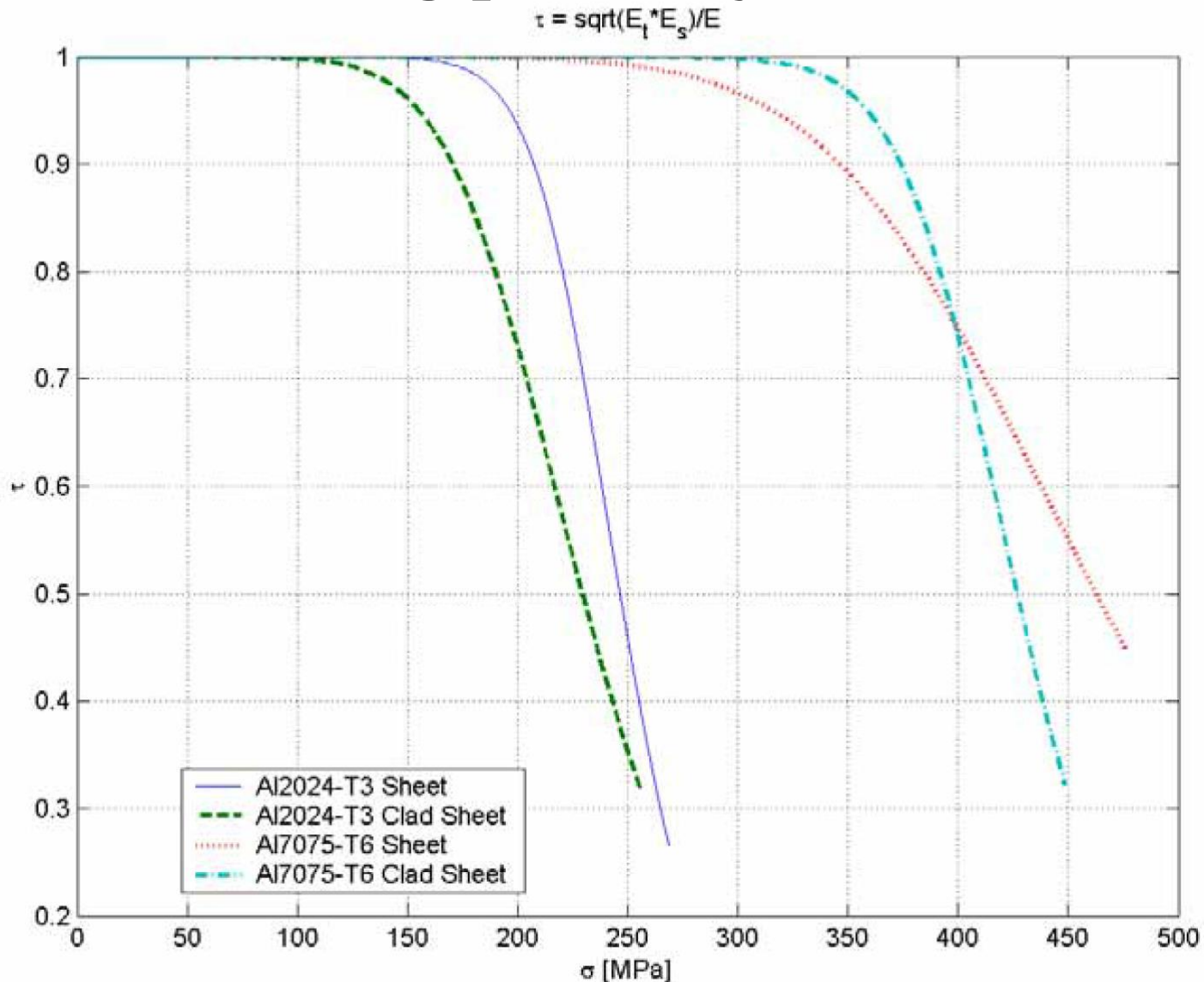
n = hardening exponent

ν = Poisson ratio

γ = density



Task 1 – Buckling plasticity correction factor



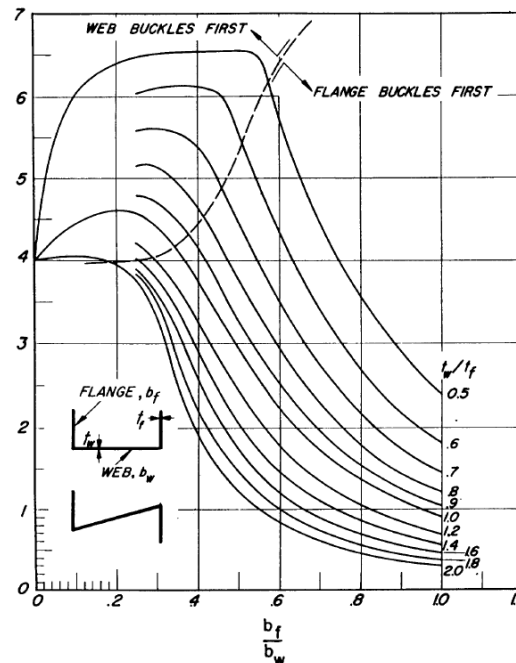
Project II: Tasks

Tasks:

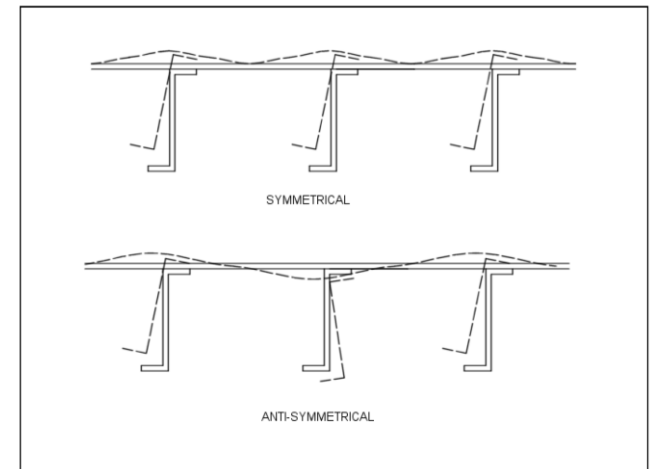
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Task 2 – Assumptions

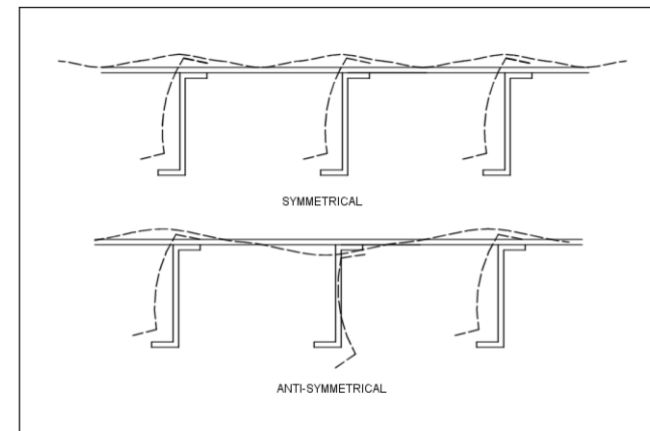
- Number of stringers sufficiently large to be considered infinitely wide panel ($n > 3/4$)
- Ribs assumed to supply a simple support
- Inter-ripped buckling and wrinkling avoided by design consideration
- For typical aerospace stringer geometry flexural-torsional coupling is small and can be neglected (pure flexural stability)
- Skin buckling coupling by coupling of stringer lateral/torsional/local buckling considered
- Decoupling between skin and flexural buckling
- Minimum weight optimum design corresponds to the condition when skin and flexural buckling occur at the same time



Flexural-torsional buckling coupling



Skin-flexural-torsional buckling coupling



Task 2 – Panel sizing equations

$$\sigma_a = \frac{N}{\bar{t}}$$

$$\sigma_{Eul} = \frac{\pi^2 \cdot c \cdot E \cdot \tau}{(L/\rho)^2} = \frac{\pi^2 \cdot E \cdot \tau}{(L'/\rho)^2}$$

$$\sigma_L = \eta \cdot k \cdot E \cdot \left(\frac{t}{b}\right)^2$$

- Lightest design is the one where local and flexural instability coincides at the highest stress

α = structural efficiency parameter

η = plate buckling plastic correction factor ($\eta = E_s/E$)

τ = column buckling plastic correction factor ($\tau = E_t/E$)

L = rib spacing

c = end fixity coefficient ($c=1$)

ρ = section gyration ratio

$$\sigma_{opt}^4 = \sigma_a^2 \cdot \sigma_{Eul} \cdot \sigma_L$$



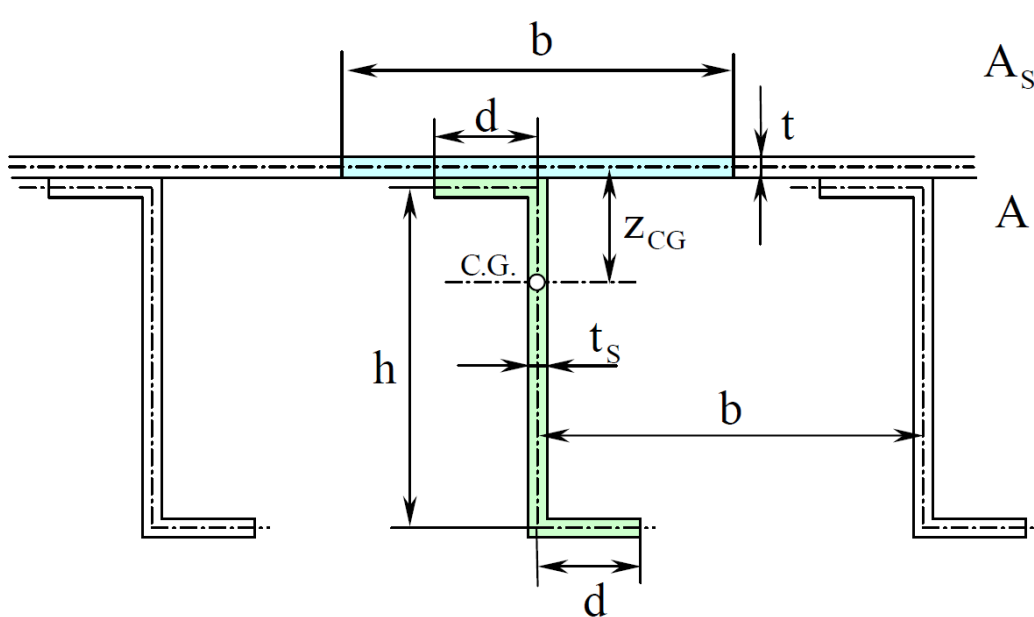
$$\sigma_{opt} = \alpha \cdot \sqrt{\frac{N \cdot E \cdot \bar{\tau}}{L'}}$$

$$L' = \frac{L}{\sqrt{c}}$$

$$\alpha = \sqrt{\pi} \cdot (k)^{0.25} \cdot \sqrt{\frac{\rho \cdot t}{b \cdot \bar{t}}}$$

$$\bar{\tau} = (\eta \cdot \tau)^{0.5} \cong \left(\frac{E_s}{E}\right) \sqrt{\left(\frac{E_T}{E_s}\right)}$$

Task 2 – Panel sizing: Z-shaped stringers



$$A_s = h \cdot t_s \cdot \left(1 + \frac{2d}{h}\right) = \text{Stringer cross section area}$$

$$A = b \cdot t \cdot \left(1 + \frac{A_s}{b \cdot t}\right) = \text{Section cross section area (stringer + skin)}$$

$$\frac{\bar{t}}{t} = \frac{A}{b \cdot t} = 1 + \frac{A_s}{b \cdot t}$$

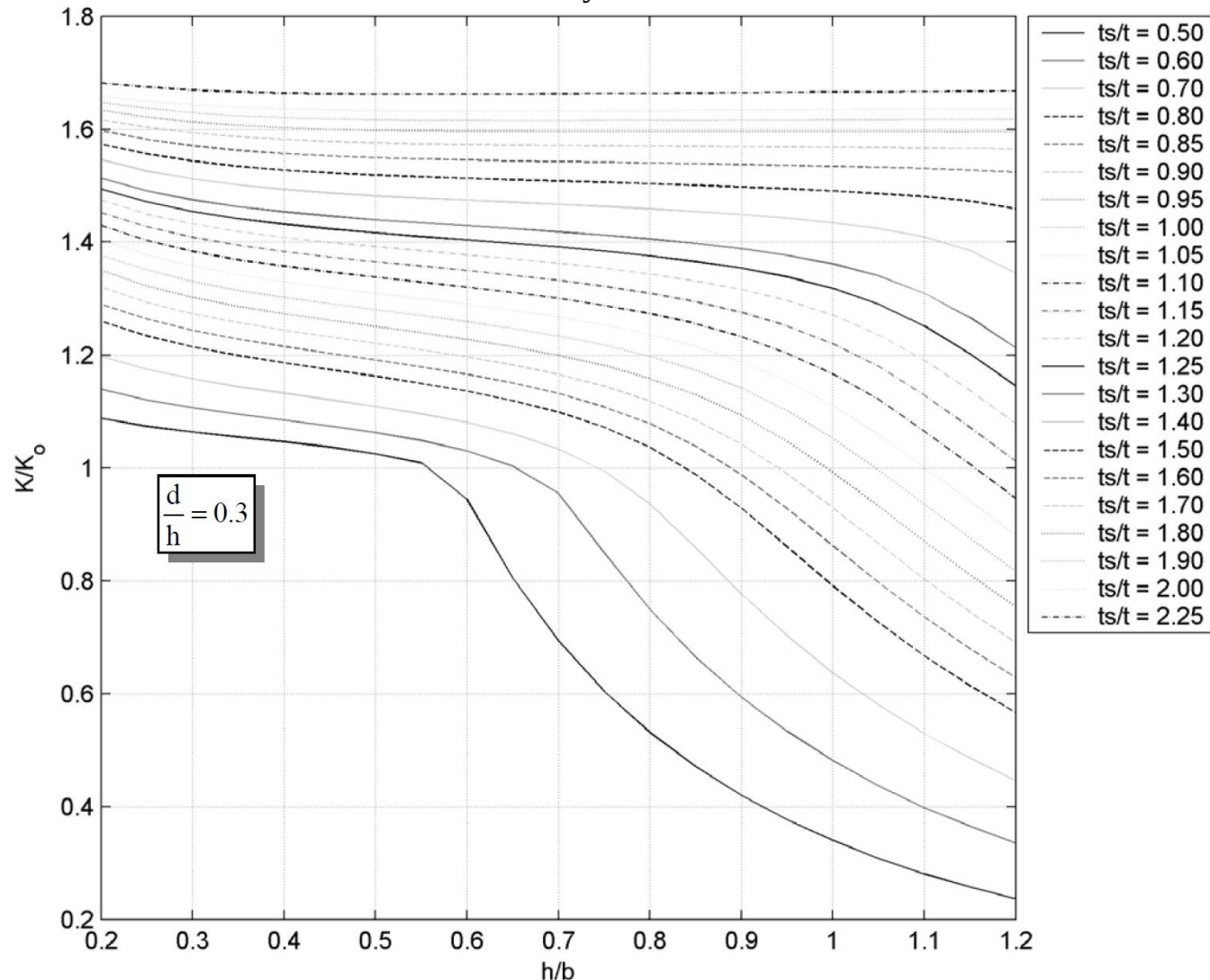
$$z_{CG} = \frac{h}{2} \frac{A_s}{A}$$

$$J_{CG} = \frac{h^2 \cdot A_s}{A/bt(1 - 2d/h)} \left[\frac{1}{3} + \frac{d}{h} + \frac{A_s}{b \cdot t} \left(\frac{1}{12} + \frac{d}{2h} \right) \right]$$

$$\rho = \sqrt{\frac{J_{CG}}{A}} = h \sqrt{\frac{A_s/bt}{(1 + A_s/bt)^2 (1 + 2d/h)} \left[\frac{1}{3} + \frac{d}{h} + \frac{A_s}{b \cdot t} \left(\frac{1}{12} + \frac{d}{2h} \right) \right]} = h \cdot f(A_s/bt ; 2d/h)$$

Task 2 – Panel sizing: Z-shaped stringers

Local instability coefficient



$$\sigma_L = k \cdot E \cdot \eta \cdot \left(\frac{t}{b} \right)^2$$

$$k = K_0 \cdot \left(\frac{K}{K_0} \right)$$

$$K_0 = \frac{4 \cdot \pi^2}{12(1-\nu^2)} \cong 3.62$$



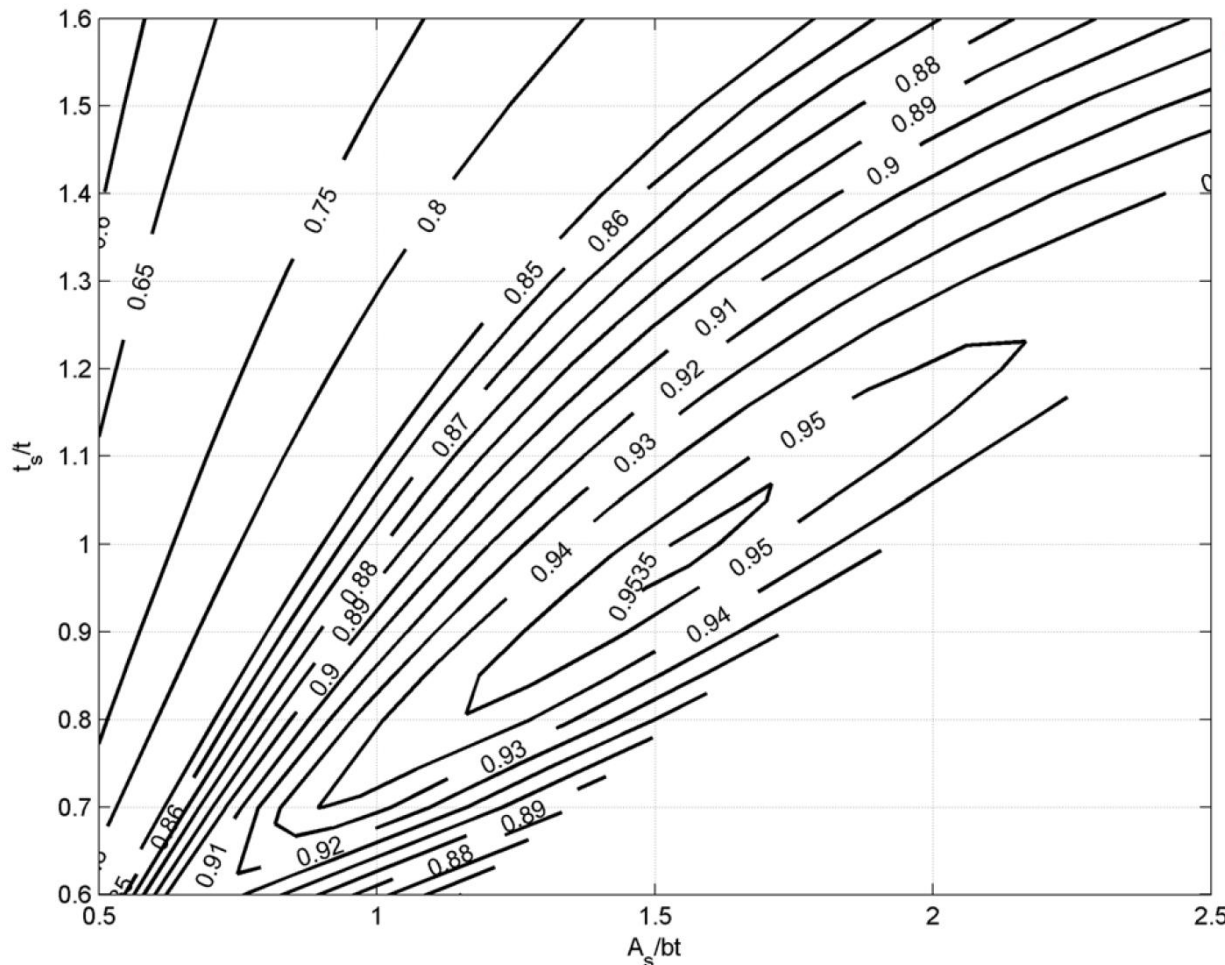
Local instability
coefficient for a simply
supported plate at all
edges ($a/b > 3$)

b = plate width

$$\frac{h}{b} = \frac{A_s / b \cdot t}{\frac{t_s}{t} \cdot \left(1 + \frac{2d}{h} \right)}$$

Task 2 – Panel sizing: Z-shaped stringers

Structural efficiency parameters



$$\sigma_{opt} = \alpha \cdot \sqrt{\frac{N \cdot E \cdot \bar{\tau}}{L'}}$$

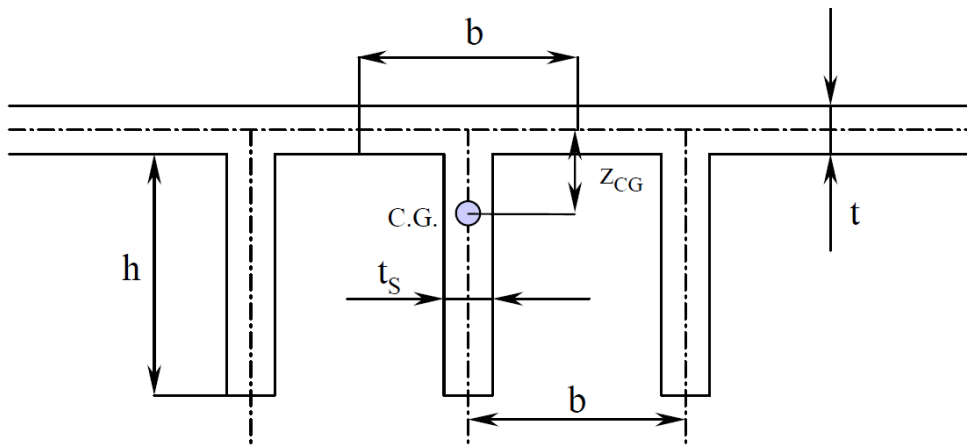
Structural efficiency
parameter

$$\alpha = \sqrt{\pi} \cdot (k)^{0.25} \cdot \sqrt{\frac{\rho \cdot t}{b \cdot \bar{t}}}$$

$$\rho = h \cdot f(A_s/bt; 2d/h)$$

$$\alpha = \sqrt{\pi} \cdot (k)^{0.25} \cdot \sqrt{f\left(\frac{A_s}{bt}; \frac{2d}{h}\right)} \cdot \sqrt{\frac{A_s/bt}{\left(1 + \frac{2d}{h}\right) \frac{t_s}{t} \left(1 + \frac{A_s}{bt}\right)}}$$

Task 2 – Panel sizing: integral stringers



$$A_s = h \cdot t_s$$

$$A = b \cdot t \cdot \left(1 + \frac{A_s}{b \cdot t} \right) = b \cdot t \cdot \left(1 + \frac{h}{b} \cdot \frac{t_s}{t} \right)$$

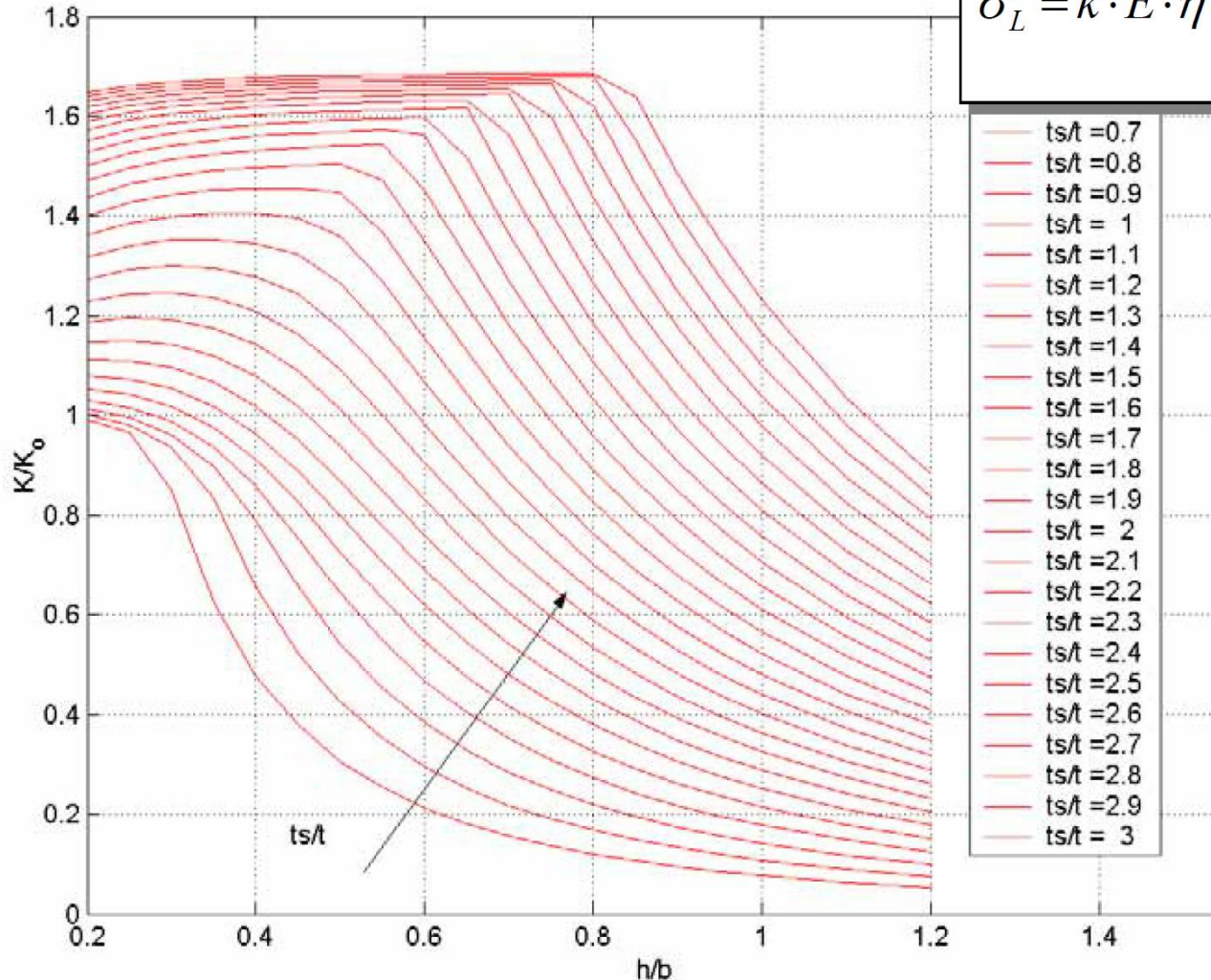
$$\frac{\bar{t}}{t} = \frac{A}{b \cdot t} = 1 + \frac{A_s}{b \cdot t} = 1 + \frac{h}{b} \cdot \frac{t_s}{t}$$

$$z_{CG} = \frac{h}{2} \frac{A_s}{A} \quad J_{CG} = b \cdot \frac{t^3}{12} + b \cdot t \cdot z_{CG}^2 + t_s \cdot \frac{h^3}{12} + t_s \cdot h \cdot \left(\frac{h}{2} - z_{CG} \right)^2$$

$$\rho = \sqrt{\frac{J_{CG}}{A}} = h \cdot \sqrt{\frac{\frac{1}{12} \cdot \frac{h}{b} \cdot \frac{t_s}{t} \cdot \left(4 + \frac{h}{b} \cdot \frac{t_s}{t} \right)}{\left(1 + \frac{h}{b} \cdot \frac{t_s}{t} \right)^2}} = h \cdot f\left(\frac{h}{b}; \frac{t_s}{t}\right) = h \cdot f\left(\frac{A_s}{b \cdot t}\right)$$

Task 2 – Panel sizing: integral stringers

Local instability coefficient



$$\sigma_L = k \cdot E \cdot \eta \cdot \left(\frac{t}{b}\right)^2 = K_0 \cdot \left(\frac{K}{K_0}\right) \cdot E \cdot \eta \cdot \left(\frac{t}{b}\right)^2$$

$$K_0 = \frac{4 \cdot \pi^2}{12(1-\nu^2)} \cong 3.62$$



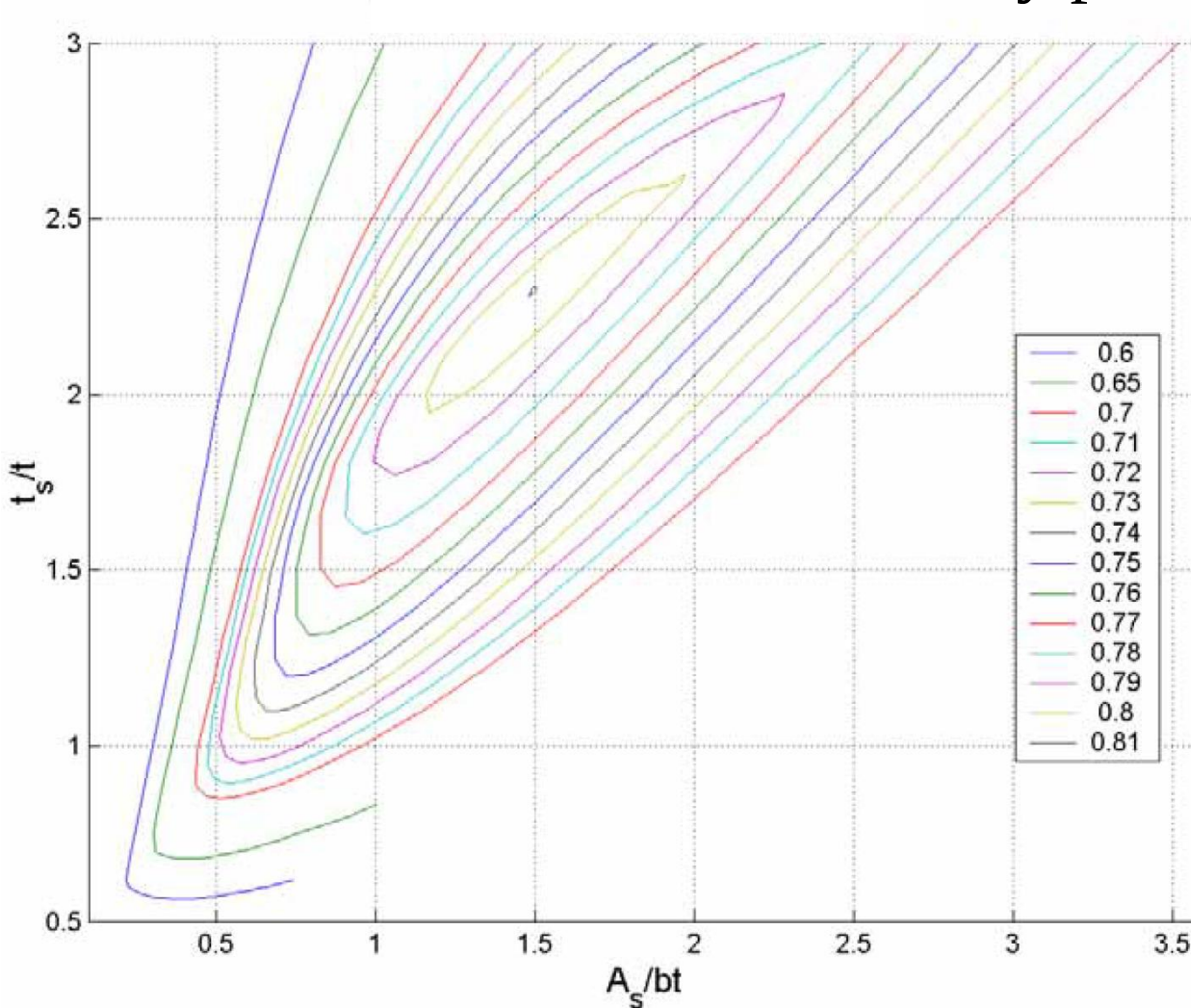
Local instability coefficient for a simply supported plate at all edges ($a/b > 3$)

b = plate width

$$\frac{h}{b} = \frac{A_s / b \cdot t}{t_s / t}$$

Task 2 – Panel sizing: Integral stringers

Structural efficiency parameters



$$\sigma_{opt} = \alpha \cdot \sqrt{\frac{N \cdot E \cdot \bar{\tau}}{L'}}$$

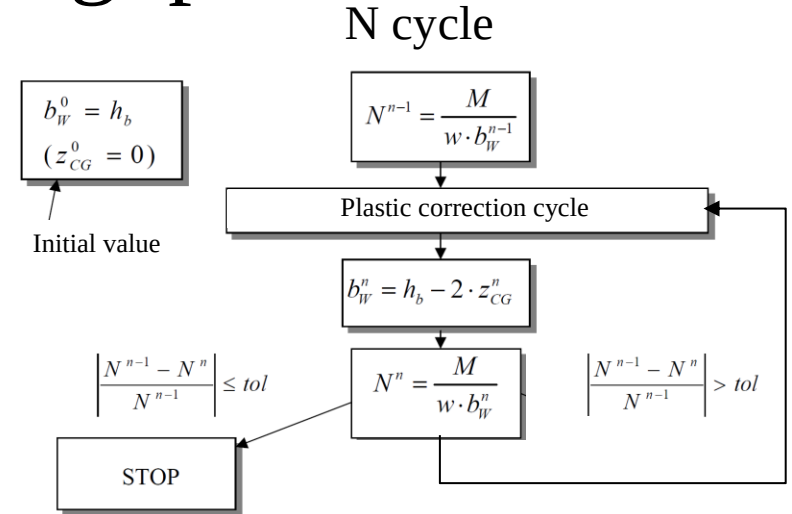
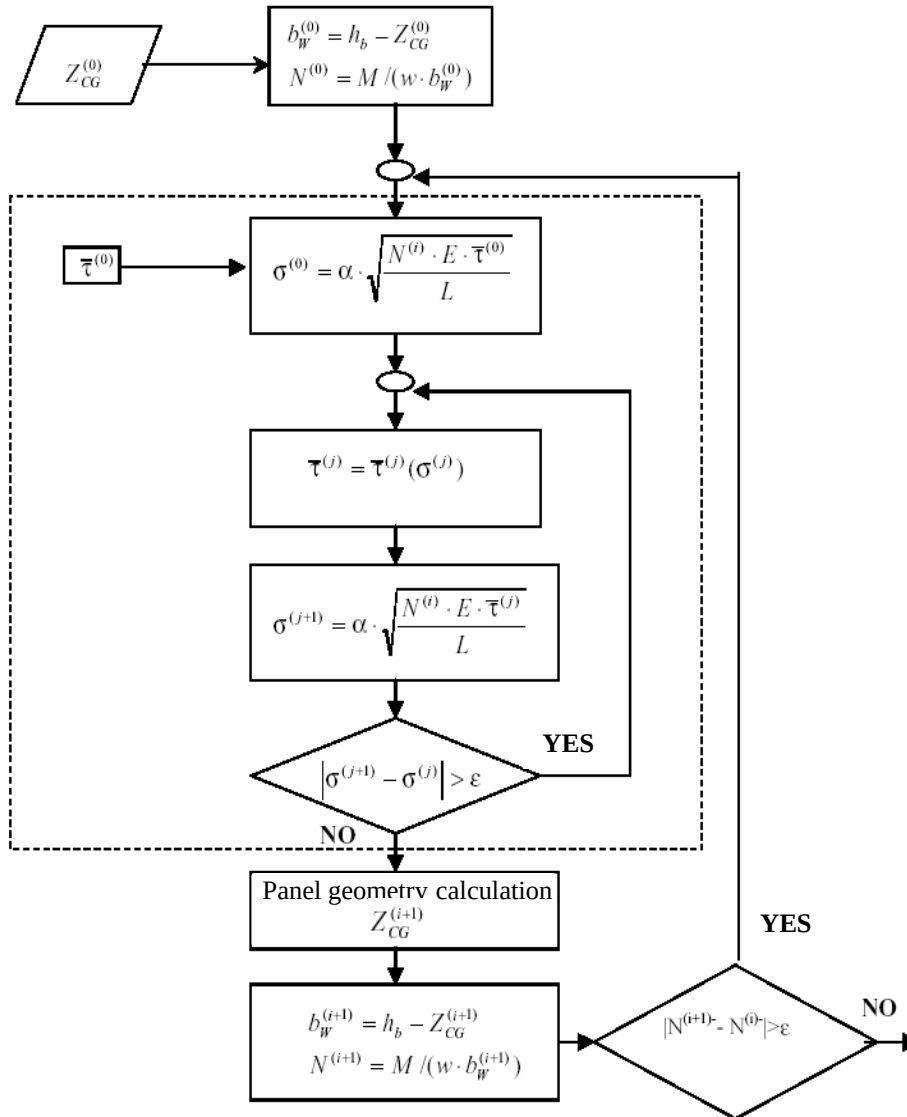
Structural efficiency
parameter

$$\alpha = \sqrt{\pi} \cdot (k)^{0.25} \cdot \sqrt{\frac{\rho \cdot t}{b \cdot \bar{t}}}$$

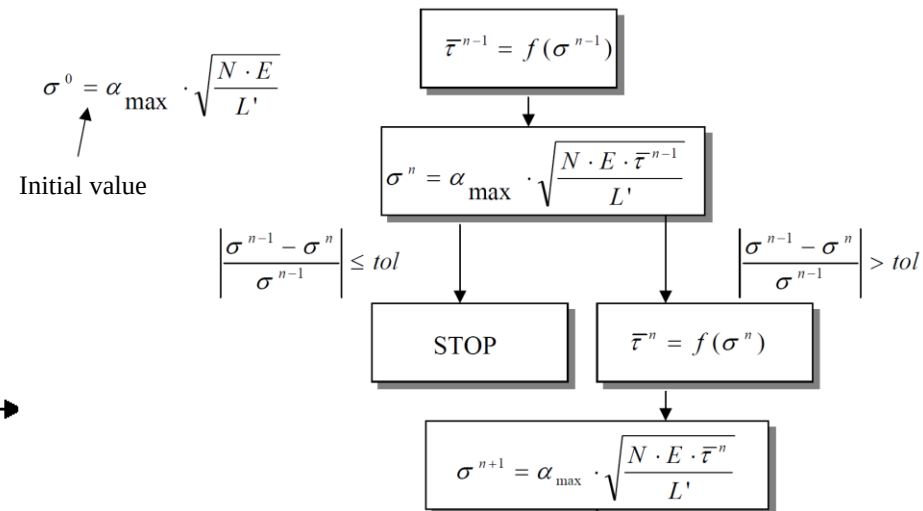
$$\rho = h \cdot f(A_s/bt)$$

$$\alpha = \sqrt{\pi} \cdot (k)^{0.25} \cdot \sqrt{f\left(\frac{A_s}{bt}\right)} \cdot \sqrt{\frac{A_s/bt}{\frac{t_s}{t} \left(1 + \frac{A_s}{bt}\right)}}$$

Task 2 – Panel sizing: procedure

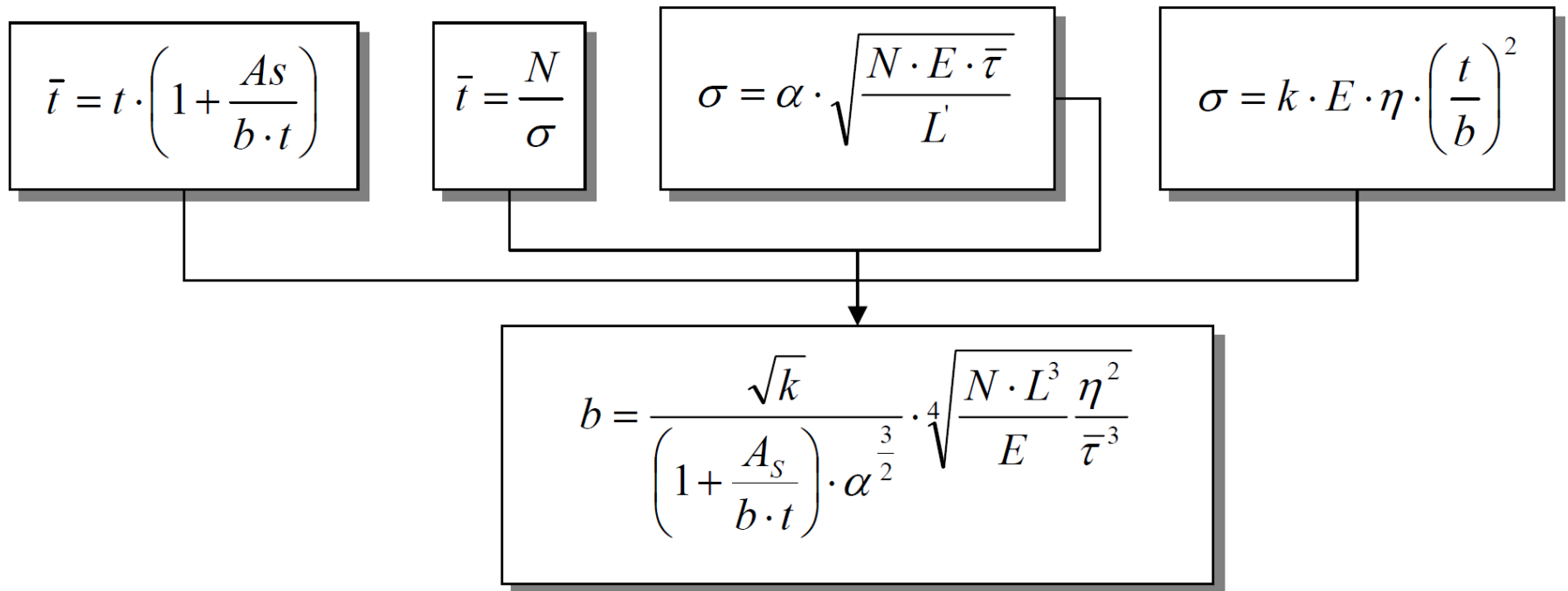


Plastic correction cycle



Task 2 – Panel sizing: procedure

Panel geometry calculation



- α_{opt} define the entire panel aspect ratio ($A_s/b \cdot t$, t_s/t , h/b)
- Calculate b
- Calculate Z_{CG}
- Calculate b_w

Project II: Tasks

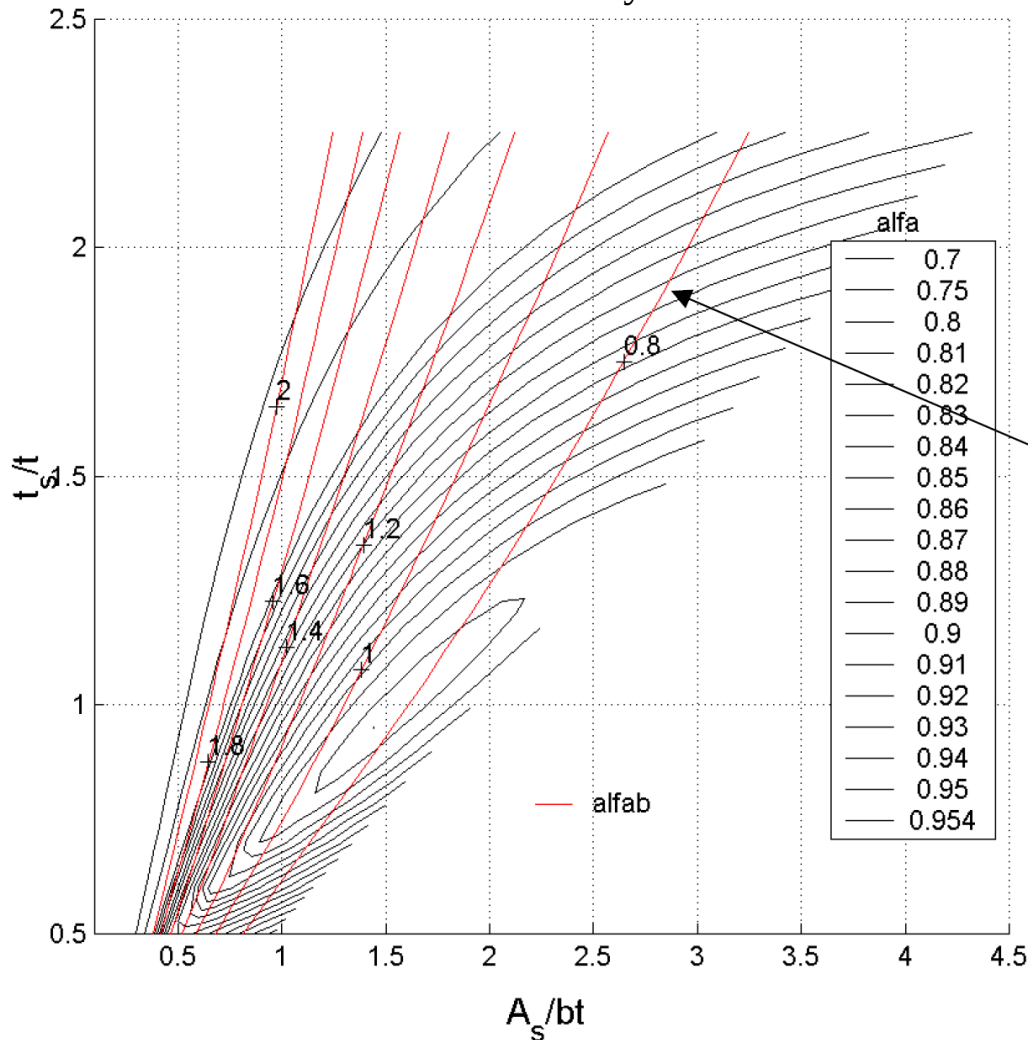
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Task 2 – Panel sizing: Z-shaped stringers

Structural efficiency parameters with manufacturing constraints

Structural efficiency fixed b



$$b = \alpha_b \cdot \sqrt[4]{\frac{N \cdot L^3}{E} \frac{\eta^2}{\bar{\tau}^3}}$$

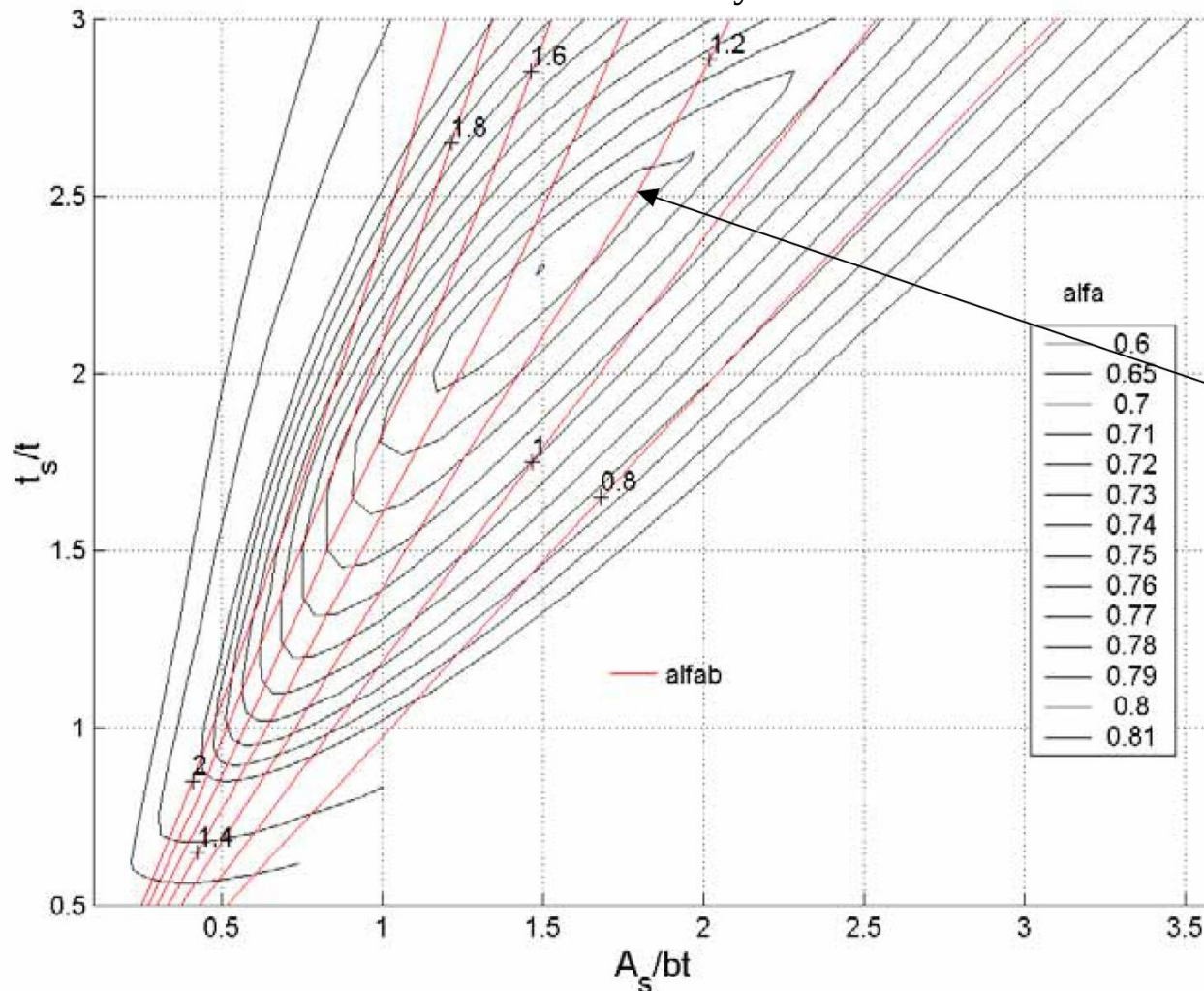
$$\alpha_b = \frac{\sqrt{k}}{\left(1 + \frac{A_s}{b \cdot t}\right) \cdot \alpha^{\frac{3}{2}}}$$

Structural efficiency parameter
for constant b

Task 2 – Panel sizing: integral stringers

Structural efficiency parameters with manufacturing constraints

Structural efficiency fixed b



$$b = \alpha_b \cdot \sqrt[4]{\frac{N \cdot L^3}{E} \frac{\eta^2}{\bar{\tau}^3}}$$

$$\alpha_b = \frac{\sqrt{k}}{\left(1 + \frac{A_s}{b \cdot t}\right) \cdot \alpha^{\frac{3}{2}}}$$

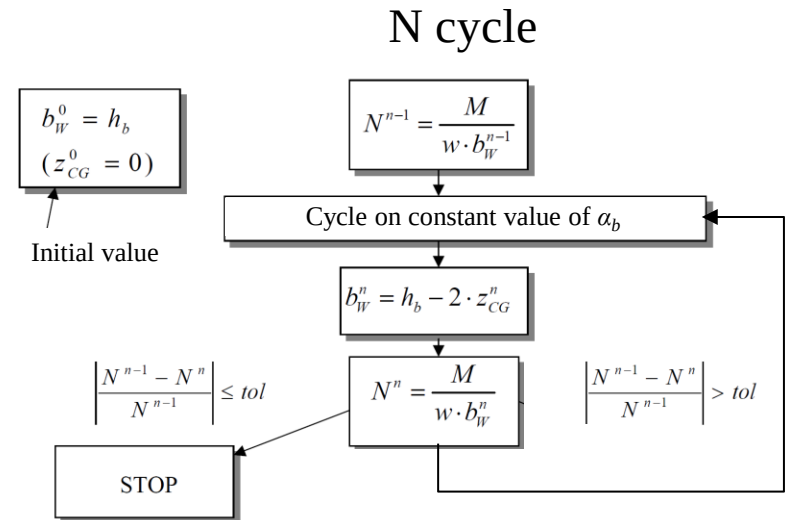
Structural efficiency
parameter for constant b

Task 2 – Panel sizing: constant b procedure

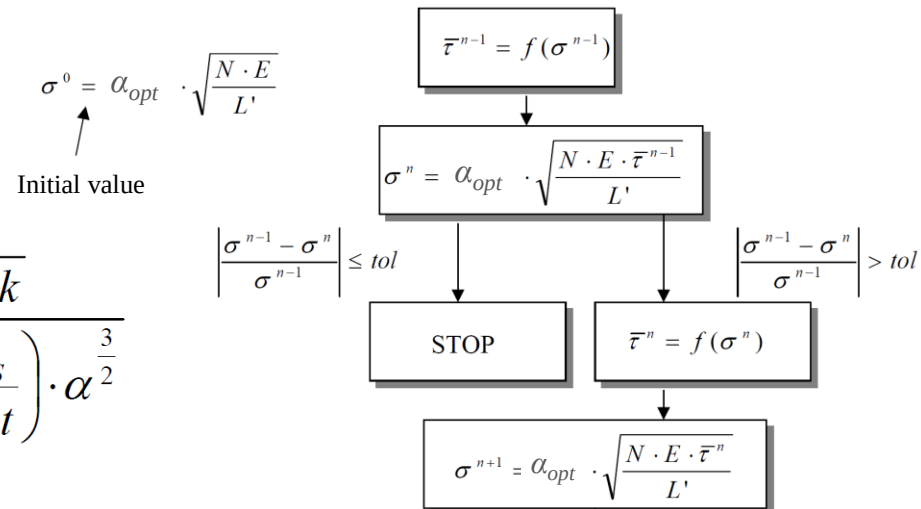
- Fix a t_s/t value
 - $0.6 \leq t_s/t \leq 1.6$ Z-shaped stringers
 - $1.0 \leq t_s/t \leq 2.5$ integral stringers
- Find for each value $A_s/(b \cdot t)$ giving α_b
- Select the couple $[t_s/t; A_s/(b \cdot t)]$ resulting in $\alpha_{\max}$
- Plasticity correction cycle is required to find the fixed value of α_b
- Iteration cycles are required to find proper α_b value as $\alpha_b = f(N)$

$$\alpha_b = \frac{b}{\sqrt[4]{\frac{N \cdot L^3 \cdot \eta^2}{E \cdot \bar{\tau}^3}}} \quad \leftarrow \text{Fixed } \alpha_b \text{ value}$$

$$\alpha_b = \frac{\sqrt{k}}{\left(1 + \frac{A_s}{b \cdot t}\right) \cdot \alpha^{\frac{3}{2}}}$$

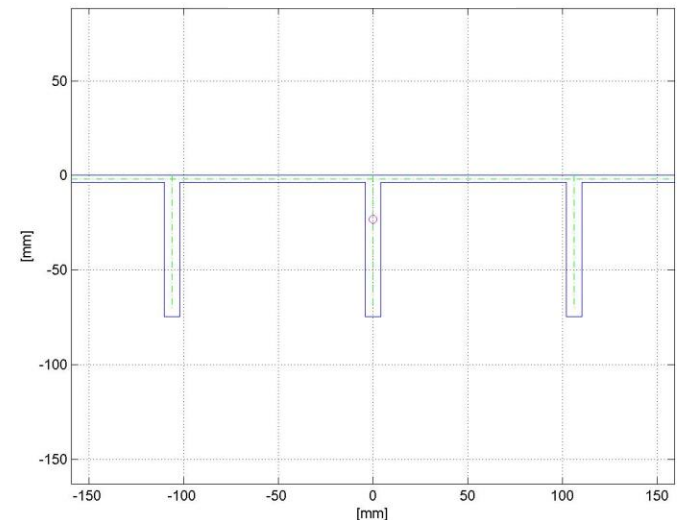
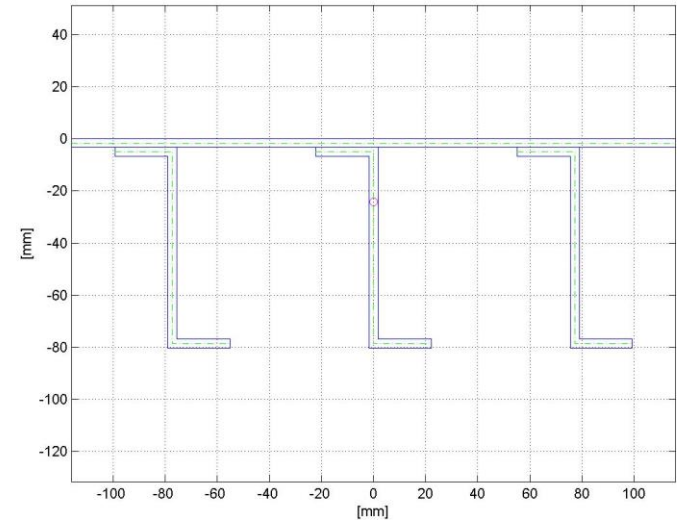


Plastic correction to determine α_b



Task 2 – Deliverable

- Select a specific rib spacing based on exiting aircraft
- Select Al 7075-T6 bare as material
- Perform the panel sizing for each of the three sections in the following conditions
 - b variable in each section
 - Z-shaped stringer
 - Integral stringers
 - b constant in each section (selected from similar aircraft)
 - Z-shaped stringer
 - Integral stringers
- Calculate optimal stress and geometry in each panel section
- Calculate the weight of each section



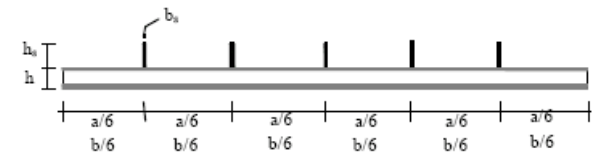
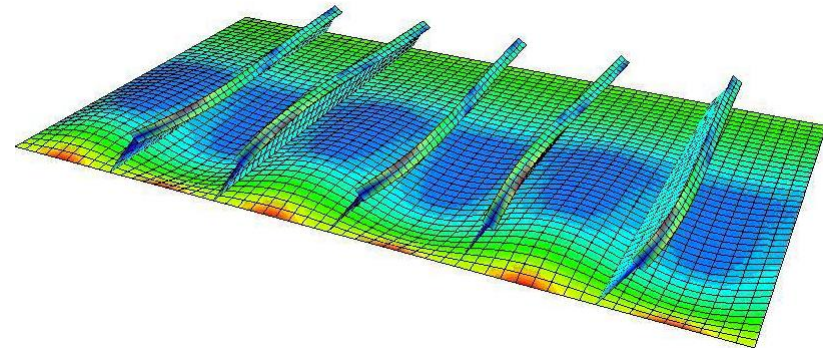
Project II: Tasks

Tasks:

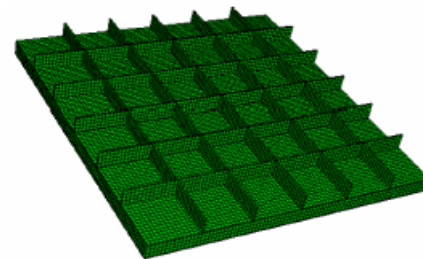
1. Preliminaries:
 - I. Retrieve the lightest wing box from Project I, and the limit condition (pull-up)
 - II. Modify the geometry to have a rectangular wing box
 - III. Material selection
2. Minimum weight sizing of stiffened panels:
 - I. Z-shaped stringers
 - II. Integral stringers
3. Minimum weight sizing of stiffened panels (manufacturing constraints):
 - I. Z-shaped stringers
 - II. Integral stringers
4. Additional considerations
 - I. Material selections
 - II. Rib optimal spacing

Task 3 – Effect of different material properties

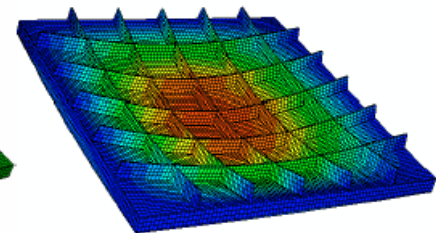
- Material σ_{tu} is not specified in the problem
- Perform a sensitivity analysis on different materials
- Perform the same analysis selecting the following materials:
 - Al 2024-T3 extruded bar
 - Al 2090-T83
 - Ti-6Al-4V sheet aged (intermediate values)
- Perform the panel sizing for root and tip sections in the following conditions
 - b variable in each section
 - Z-shaped stringer
 - Integral stringers
 - b constant in each section (selected from similar aircraft)
 - Z-shaped stringer
 - Integral stringers
- Calculate optimal stress and geometry in each panel section
- Calculate the weight of each section
- Compare the materials (including 7075)



(a)



(b)

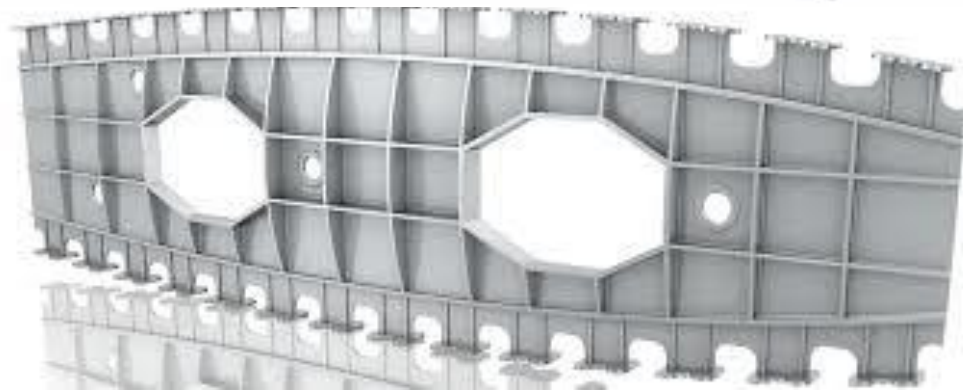
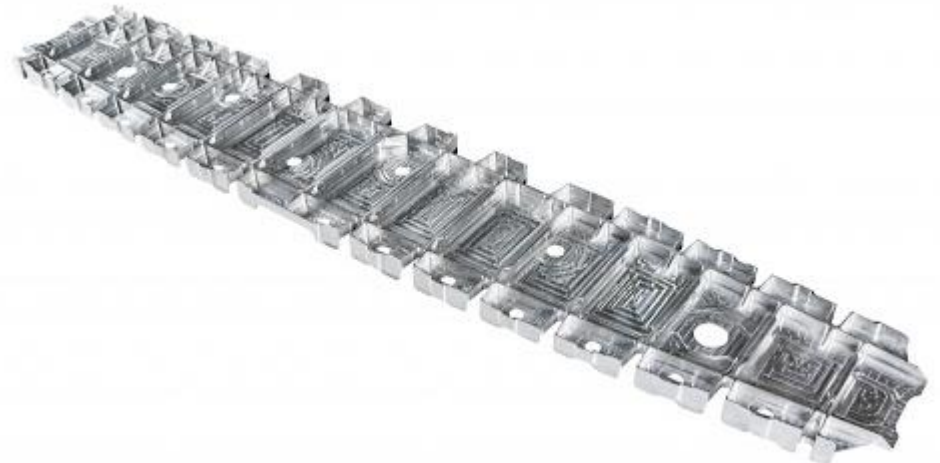


(c)

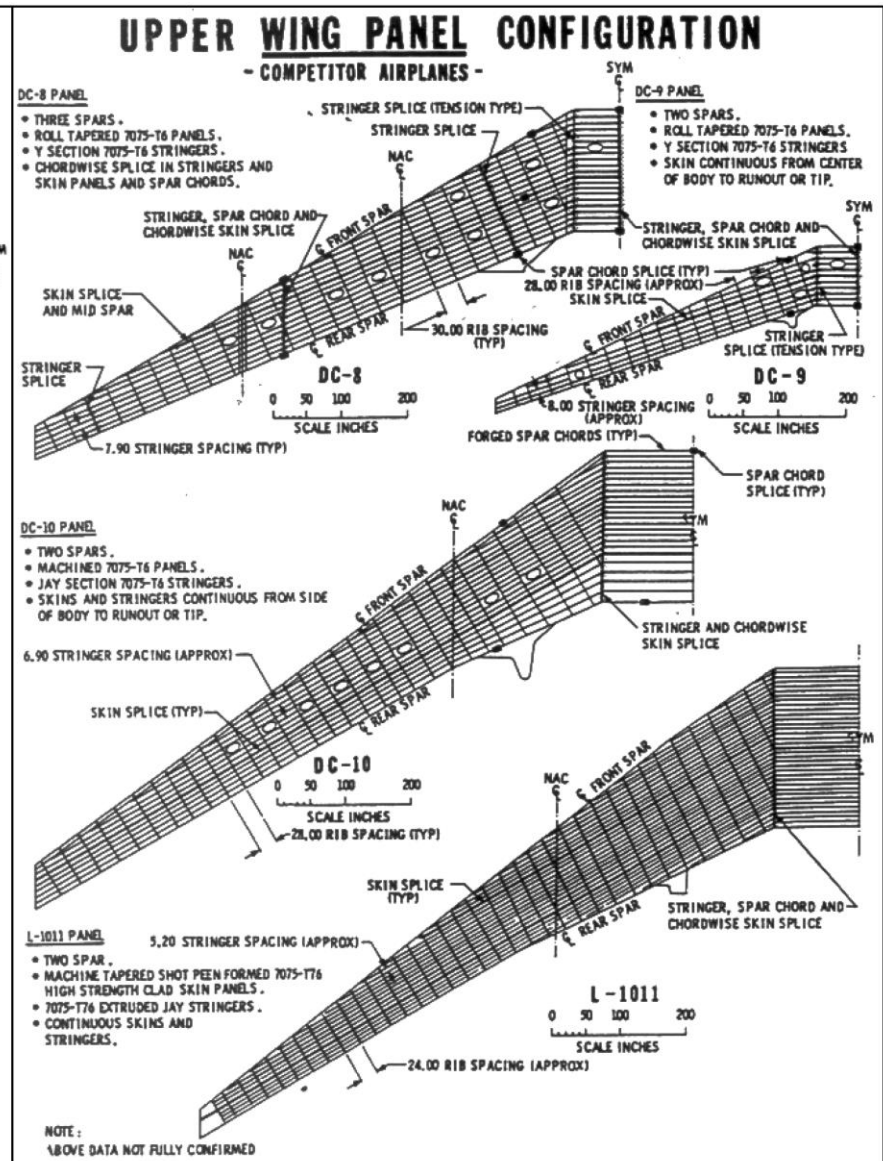
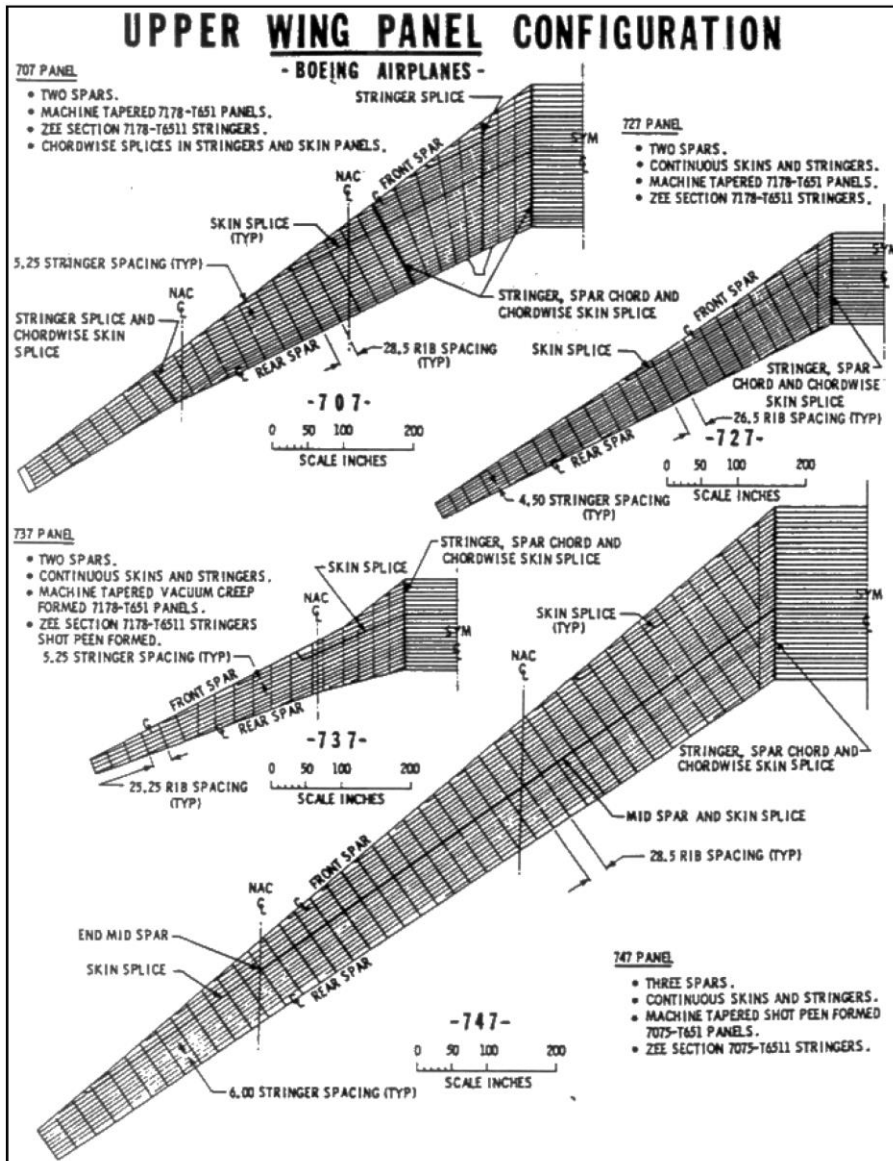
Task 3 – Rib optimal spacing

Optional

- The rib spacing L was fixed
- Assuming:
 - h_b = rib height
 - t_{rib} = average rib thickness
- Select Al 7075-T6 for both ribs and skins
- Calculated the minimum weight rib spacing in case:
 - Purely elastic stress
 - Elastic-plastic stress
- Deliver a section describing the methodology used to calculate the value



Project II: wing examples



Approach

The project is a learning process which allows you to “study” the theory and apply it in a design which tries to “mimic” real life working situation [transfer of theoretical notions to practical situations].



Project II: Bibliography

- **G. Gerard**, “Minimumweight analysis of compression structures”, New York University Press, 1956
- **D. J. Farrar**, “The design of compression structures for minimumweight”, Journal of the Royal Aeronautical Society, pp. 1041-1052, Nov. 1949
- **E. J. Catchpole**, “The optimum design of compression surfaces having unflanged integral stiffeners”, Journal of the Royal Aeronautical Society vol. 58 pp. 765-768, Nov. 1954
- **E. F. Bruhn**, “Analysis and design of flight vehicle structures”. Tri-State Offset, 1965